RoboBIM: From Coverage to Correctness in Building-Information Priors for Robot Navigation

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Abstract— Building Information Models (BIM), encoded as Industry Foundation Classes (IFC) files, carry architectural priors that could let an indoor robot skip minutes of onboard metric-semantic mapping. In our workshop system, RoboBIM [1], we showed that a staged LLM agent can populate room-type, hazard, and traversability fields that a deterministic IFC parser leaves empty. That study reported coverage on four buildings from single, unreplicated runs, against one weak deterministic strawman – leaving the two questions any reviewer asks hardest unanswered: what does the model buy over a strong deterministic baseline, and are its outputs correct, not merely non-null? This paper answers both. We build a benchmark of 150 procedurally generated buildings whose ground-truth connectivity is verified by a raster flood-fill cross-check independent of the scoring pipeline (2,769/2,769 pairs agree), and show our topology-recovery method beats three deterministic ablations and an independently published external tool at $p < 10^{-6}$ (paired Wilcoxon, $n = 150$), degrading gracefully rather than catastrophically under four levels of graduated IFC damage. We replace coverage with held-out accuracy for semantic labelling – leave-one-out masking of real, architect-assigned IFC labels – diagnose a systematic taxonomy failure rather than reporting it as an unexplained limitation, and fix the diagnosable part of it. Finally, we introduce a goal-resolution metric that closes the loop end to end: does a language query such as "go to the kitchen" retrieve the correct room, and can a specific robot embodiment then reach it. On 25 real buildings and 593 held-out real rooms, resolution succeeds in 51.9% of cases and tracks label accuracy class-for-class, showing the two stages compose rather than compensate for one another. We release the generator, the verified benchmark, the annotation tool, and the full evaluation harness.

I. Introduction

Mobile robots deployed in structured indoor environments routinely reconstruct metric-semantic representations from onboard sensing – Kimera [2] and Hydra [3] are exemplars – even when a design-time digital twin of the deployment site already exists as an IFC model [4]–[6]. This duplication discards a prior the robot would otherwise spend minutes re-discovering. But the discarding is not entirely unreasonable: exported IFC is a drawing-production artefact, not a navigation one. The relationships a robot needs – which rooms are adjacent, which door joins which pair of rooms, what a room is for – are precisely the relationships that authoring tools have no reason to state, because no architectural drawing requires them.

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What our workshop paper established, and what it left open. RoboBIM [1] framed IFC-to-prior compilation as a deterministic geometric backbone plus a staged, tool-augmented LLM agent that enriches it with room types, hazards, materials, and traversability, gated by a check that rejects any agent edit which reduces connected-transition count below the deterministic baseline. On four public buildings the agent raised room-type coverage from 51/131 to 89/131 rooms and populated fields the baseline left uniformly null. This established that an agentic layer can add semantic content IFC does not state. It did not establish that the content is correct: coverage counts non-null fields, not right ones, and a system that always outputs something scores full coverage regardless of accuracy. It also compared against a single regex-and-property-set baseline – exactly the strawman that leaves "what does the LLM actually buy you" unanswered – and reported single, unreplicated runs on four buildings, which is not a foundation for a significance claim of any kind.

This paper rebuilds the evaluation, not just the pipeline. We keep RoboBIM's two-layer framing – a deterministic backbone that is trustworthy by construction, enriched by a layer that is not – but replace every measurement that could not previously distinguish "populated" from "correct." Concretely, we contribute:

1) A procedurally generated benchmark of 150 buildings whose ground-truth room connectivity is checked by a second, independent mechanism (raster flood-fill) rather than assumed correct because a generator wrote it, closing the single largest gap our own prior evaluation had: no reference to measure against at all.
2) A statistically grounded comparison – paired Wilcoxon significance and win/loss records over $n = 150$ buildings, not $n = 1$ – against three deterministic ablations and an independently published, publicly available external tool [7], replacing the single weak strawman of the workshop study.
3) A graduated damage model (four levels, from space-boundary stripping to corrupted units) that isolates room recovery as the binding constraint under realistic IFC corruption, in place of an untested claim that the pipeline is robust.

4) A correctness metric for semantic labelling – leave-one-out masking of real, architect-assigned room names – that replaces coverage outright, together with a diagnosed and partially repaired taxonomy failure reported with its root cause rather than left as an unexplained limitation.
5) A goal-resolution metric that closes the loop the whole system exists to serve: whether a language query resolves to the correct room and whether a specific robot embodiment can then reach it, evaluated on 25 real buildings and 593 held-out real rooms.

Section II positions this work against sensor-driven scene graphs, open-vocabulary and language-grounded graphs, BIM/IFC as a structured prior, and the space-boundary and room-recovery literature that neither our workshop paper nor the surveyed BIM-to-graph work engages with, despite solving exactly the problem that blocks it.

II. Related Work

Sensor-driven 3D scene graphs. Armeni et al. [8] introduced the layered formulation – objects nested in rooms nested in a building – that nearly all later work inherits. Kimera [2] made construction fully automatic from visual-inertial data, deriving places from a Generalised Voronoi Diagram of free space and segmenting rooms at the doorway constrictions between Voronoi regions; Hydra [3] made this real-time and loop-closure aware, and Mono-Hydra [9] showed it degrades gracefully to monocular input. S-Graphs [10], [11] instead makes rooms and walls factors in the SLAM optimisation itself, and its architectural-plan variants [12], [13] use a BIM-derived graph as a registration target for localisation – close to our problem, but silent on what to do when the BIM's own topology is missing, which is the case we address. Every system in this cluster estimates its hierarchy from noisy sensing; ours reads the outer layers (building, storeys, exact element geometry) directly from IFC and estimates only the middle layer – rooms and their connectivity – that IFC omits.

Open-vocabulary and language-grounded graphs. ConceptGraphs [14] and HOV-SG [15] attach open-vocabulary, CLIP-grounded features to segments recovered from RGB-D streams, and Clio [16] prunes the resulting graph against a task list rather than materialising everything a sensor can see. SayPlan [17] demonstrated the property our own goal-resolution and navigation stages depend on: an LLM can plan over a building-scale graph if permitted to collapse and expand it rather than ingest the whole scene at once, and EmbodiedRAG [18] and Taskography [19] show retrieval- and graph-scale considerations matter as much as the underlying representation. These methods solve semantic richness without a prior map; we solve it with one, which changes what is worth estimating and what is worth verifying.

BIM and IFC as a structured prior. The AEC community arrived at graph representations independently. ifcOWL [20] mechanically maps the IFC schema to OWL, preserving everything at the cost of a graph that is awkward to query; the Building Topology Ontology [21] responds with a deliberately minimal core closer to what a robot needs – completeness and queryability are in tension, and robotics wants the queryable end. For navigation specifically, Zhu et al.'s IFC-Graph [22] connects neighbouring spaces through doors for indoor pathfinding, and Gopee et al. [23] and Vega Torres et al. [24] export navigation maps and localise a robot against a BIM-derived map, respectively; Pix2G [25] runs the inverse direction in real time, producing BIM-like maps from onboard sensing. Every one of these presumes a well-formed spatial hierarchy already exists in the file – exactly the presumption our own IFC audit (Sec. II, and detailed in our extended technical report) shows fails on realistic exports: zero IfcRelSpaceBoundary instances across 257 MB of real IFC, and only 51% of storeys containing any IfcSpace at all. Coupling IFC to language models is more recent: Iranmanesh et al. [26] retrieve grounded facts from an IFC-derived knowledge graph via LLM-emitted Cypher rather than trusting parametric memory, and Text2BIM [27] inverts the direction, generating models from text. Most directly, BIM-Edit [28] benchmarks LLMs on natural-language editing of IFC across 324 tasks and finds the best model reaches only 49.5% average score, with topological instructions the weakest category – a result that argues, independently of our own findings, that raw IFC is a poor interface for a language model and that an explicit, verified topological graph is the right intermediate representation. Our own prior system [1] sits in this cluster: it established that a staged agent, gated against topology regression, can enrich such a graph, but measured only whether fields were populated, not whether they were right, and compared against a single deterministic prefilter rather than the field's actual state of the art.

Recovering spaces and boundaries. A literature largely unknown in robotics has, in fact, long known IFC space data is unreliable – it surfaced first in building-energy simulation, which needs second-level space boundaries for thermal analysis. Lilis et al. [29] generate that topology from raw IFC geometry, explicitly handling clash errors and malformed space volumes, and Fichter et al. [30] note plainly that many delivered files lack IfcSpace entirely and generate the geometry themselves. Where spaces are absent altogether, room recovery becomes an occupancy-segmentation problem: Bormann et al. [31] survey the morphological, distance-transform, Voronoi, and feature-based families that this literature and the sensor-driven scene-graph literature both converge on, and Ochmann et al. [32] reconstruct volumetric building models from point clouds via graph-cut energy minimisation over wall centre-lines – solving the same double-sided-wall pathology that any IFC-derived recov-

TABLE I: Positioning. Ground-truth verification, statistical power, and correctness (vs. coverage) measurement are what distinguish this work from its own predecessor and from external IFC-to-graph tools.

Property	RoboBIM WS [1]	External tools	This work
IFC as prior source	✓	✓	✓
No perception required	✓	✓	✓
Independently verified GT	✗	✗	✓
Statistically significant (n>1)	✗	✗	✓
Correctness, not coverage	✗	partial	✓
Graduated robustness evaluation	✗	✗	✓
End-to-end goal resolution	partial	✗	✓

ery method must also handle, since a wall is a solid with two faces rather than a zero-thickness line. Kim and Min [33] add an LLM layer on top of geometric segmentation to assign functional labels, which is the same task our held-out labelling evaluation targets, but without a correctness measurement against ground truth. This cluster solves precisely the problem that blocks reliable BIM-to-graph construction, and none of the BIM-to-robotics work surveyed above cites it.

Affordance- and capability-aware planning. SayCan [34] and AutoGPT+P [35] ground LLM plans in robot affordances and capabilities once a semantic prior is available; our navigation stage is designed as a drop-in producer for such planners, parameterised per embodiment (wheeled, legged, human) rather than assuming a single robot.

Positioning. This paper differs from the sensor-driven literature in that it never estimates geometry – IFC states it exactly – and from prior BIM-to-robotics work, including our own [1], in three concrete ways that follow directly from the gaps above: ground truth is independently verified rather than assumed, correctness is measured by holding out real labels rather than counting non-null fields, and the evaluation is powered by $n = 150$ synthetic buildings with statistical significance testing rather than $n = 1$ single-run measurements on four. Table I summarises this against the workshop study and the closest external systems.

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